

Homework 3 - BLAS Practice

In this homework, you are required to implement a simple algorithm in C language and compare the performance of different BLAS implementations on your program. Please consider **at least four** of the following BLAS libraries in your report:

- The Netlib BLAS/CBLAS
- Intel oneMKL
- AMD BLIS
- GotoBLAS

See the NY Times article.

This GotoBlas is not actively maintained now, so it's unclear if the code can be easily used or not.

- ATLAS (Its GEMM is used from Kazushige Goto's)
- OpenBLAS (An optimized BLAS library based on GotoBLAS2 1.13 BSD version.)
- Any other implementation listed on the wiki page.

The Steepest Descent Method

The **steepest descent** is an algorithm for finding a local minimum or minimizing a given function. You are required to write a program with the steps:

- (1) Generate or read a **symmetric positive-definite (SPD)** matrix A and vector b .
- (2) Apply Algorithm 1.
- (3) Print the solution of $Ax = b$.

Algorithm 1 Steepest Descent method

Require: The input SPD matrix $A \in \mathbb{R}^{d \times d}$, vector $b \in \mathbb{R}^d$, and a small tolerance $\epsilon_{\text{tol}} \in \mathbb{R}$.
 $x \leftarrow [0 \ 0 \ \dots \ 0] \in \mathbb{R}^d$
 $\epsilon \leftarrow \infty$
 $r \leftarrow b$
while $\epsilon > \epsilon_{\text{tol}}$ **do**
 $\alpha \leftarrow \frac{r^T r}{r^T A r}$
 $x \leftarrow x + \alpha r$
 $r \leftarrow b - Ax$
 $\epsilon \leftarrow \|r\|_2$
end while
return x $\triangleright$ solution of $Ax = b$

Report Requirement

1. Implement the steepest descent method in C and link it with different BLAS implementations that you choose. Note that you should utilize the functions provided by BLAS when doing matrix, vector computations and data copy. Also, you are required to **include your code in the report pdf file**.
Tip: Most implementation (including Intel MKL, AMD BLIS) supports the CBLAS interface. Therefore, you only need to write one version of code and link it with different implementations.
2. Test the algorithm with a SPD matrix. Check the solution of $Ax = b$ given by your algorithm against the solution calculated by SciPy or MATLAB to verify the correctness of the algorithm.
3. Compare and discuss the performance of different implementations of BLAS with matrices on sizes of 100x100, 500x500, 1000x1000 and 4000x4000. Are some of the implementations faster? Why? You can cite resources or perform extra experiments to strengthen your argument.
4. You should specify whether your implementation has multi-thread enabled. Furthermore, please provide information of your environment and hardware specifications.